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TOKEN TRACKING SYSTEM

Abstract

An apparatus, method, and machine-readable medium for tracking gaming tokens, such as chips, tiles, etc. within a tray are disclosed. The tray features a token support structure (e.g., a sloped column or semi-cylindrical slot for stacking circular articles) with internal light sensors. A tracking controller detects ambient light via these sensors to determine token count. A machine learning model analyzes image data of gaming-token features, such as token edge patterns, to identify primary and minority (misplaced) token patterns, determining specific denominations for both. Information about minority tokens, including their denomination, is indicated to electronic devices. Instructions for these operations are storable on non-transitory machine-readable mediums. The system can further compute an overall value for a grouping (e.g., a stack) of tokens in the token support structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/836,668, filed Jun. 9, 2022 (hereinafter “the '668 application”). The '668 application claims the benefit of U.S. Provisional Patent Application No. 63/212,771, filed Jun. 21, 2021. The entire disclosures of U.S. patent application Ser. No. 17/836,668 and U.S. Provisional Patent Application No. 63/212,771 are hereby incorporated by reference herein.

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FIELD OF THE INVENTION

[0003] The present invention relates generally to gaming systems, apparatus, and methods and, more particularly, to tracking of physical objects in a gaming environment.

BACKGROUND

[0004] Casino gaming environments are dynamic environments in which people, such as players, casino patrons, casino staff, etc., take actions that affect the state of the gaming environment, the state of players, etc. For example, a player may use one or more physical tokens to place wagers on the wagering game. A player may perform hand gestures to perform gaming actions and/or to communicate instructions during a game, such as making gestures to hit, stand, fold, etc. Further, a player may move physical cards, dice, gaming props, etc. A multitude of other actions and events may occur at any given time. To effectively manage such a dynamic environment, the casino operators may employ one or more tracking systems or techniques to monitor aspects of the casino gaming environment, such as credit balance, player account information, player movements, game play events, and the like.

[0005] However, one challenge to such tracking systems is tracking the complexity of the system elements, particularly regarding the tracking of money. For example, multiple cameras at, or around, a gaming table may take pictures of casino tokens (e.g., casino chips) at a gaming table from different perspectives (i.e., from the perspective of the camera lenses). A computer then analyses the images to detect the differences between visible features of objects in the images. However, certain image characteristics (e.g., colors, object outlines, etc.) can appear different from image to image based on slight differences in environmental lighting and/or based on in changes of perspective of the cameras used to take the images of the objects. Consequently, some contemporary systems fail to identify some objects. Others require significant computing resources to capture highly detailed images of chips sufficient to identify the visible features needed to determine chip values.

[0006] Accordingly, a new tracking system that is adaptable to the challenges of dynamic casino gaming environments is desired.

SUMMARY

[0007] According to aspects of the present disclosure, an apparatus and a method provide for tracking gaming tokens. The approach features a token tray including a token support structure, the token support structure having a structure that positions a group of gaming tokens into an orientation from which a respective edge of each gaming token can be viewed by an image sensor. A plurality of light sensors are positioned inside the token support structure. A tracking controller performs operations, or the method includes operations, to: detect a level of ambient light at the light sensors; determine, based on the detected light, a number of gaming tokens in the token support structure; identify, using a machine learning model analyzing image data of gaming token edges, a first pattern on majority gaming tokens and a second, different pattern on any minority gaming tokens; determine a primary token denomination value from the first pattern associated with the token support structure; determine a specific denomination value for any minority gaming token from the second pattern; and indicate, to one or more electronic devices, details of any minority gaming token including its specific denomination being different from the primary token denomination. Additionally, according to aspects of the present disclosure, one or more non-transitory machine-readable mediums have instructions stored thereon, which when executed by a set of one or more processors of an apparatus cause the apparatus to perform operations including: detecting a level of ambient light at each of a plurality of light sensors of a chip tray, wherein the plurality of light sensors are positioned inside a column of the chip tray; determining, in response to detection of the level of ambient light at each of the plurality of light sensors, a number of gaming chips placed inside the column; identifying, by a machine learning model via electronic analysis of image data of the gaming chips inside the column, a first pattern on the edge of at least a relative majority of the gaming chips, and a second, different pattern on at least one minority gaming chip; determining, based on the first pattern, a primary chip denomination value associated with the column; determining, by the machine learning model based on the second pattern, a specific denomination value for the at least one minority gaming chip; and indicating, via an electronic communication to one or more electronic devices, the presence of the at least one minority gaming chip along with its specific denomination value being different from the primary chip denomination. In specific embodiments, the token support structure (or column) includes a sloped column or a semi-cylindrical slot configured to group circular-shaped gaming tokens (or chips) into a stack. The disclosed invention may be further enhanced by features such as specific sensor types, controlled light emitters, calibration modules, diagnostic modules, advanced machine learning logic, distributed processing architectures, and computation and indication of a token-grouping value (or chip-column value) accounting for all identified denominations.

[0008] Additional aspects of the invention will be apparent to those of ordinary skill in the art in view of the detailed description of various embodiments, which is made with reference to the drawings, a brief description of which is provided below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram of an example gaming system with a chip tray according to one or more embodiments of the present disclosure.

[0010] FIG. 2 is an architectural diagram of an exemplary gaming system having a chip tray according to one or more embodiments of the present disclosure.

[0011] FIG. 3 is a diagram of an exemplary sensor system for a chip tray according to one or more embodiments of the present disclosure.

[0012] FIG. 4A, 4B, and 4C are diagrams that illustrate different perspective views of a column of a chip tray according to one or more embodiments of the present disclosure.

[0013] FIG. 5 is a diagram of an exemplary chip tray according to one or more embodiments of the

present disclosure.

[0014] FIG. **6** is a flow diagram of an example method according to one or more embodiments of the present disclosure.

[0015] FIG. **7** is a perspective view of a gaming table configured for implementation of embodiments of wagering games in accordance with this disclosure.

[0016] FIG. **8** is a perspective view of an individual electronic gaming device configured for implementation of embodiments of wagering games in accordance with this disclosure.

[0017] FIG. **9** is a top view of a table configured for implementation of embodiments of wagering games in accordance with this disclosure.

[0018] FIG. **10** is a perspective view of another embodiment of a table configured for implementation of embodiments of wagering games in accordance with this disclosure, wherein the implementation includes a virtual dealer.

[0019] FIG. **11** is a schematic block diagram of a gaming system for implementing embodiments of wagering games in accordance with this disclosure.

[0020] FIG. **12** is a schematic block diagram of a gaming system for implementing embodiments of wagering games including a live dealer feed.

[0021] FIG. **13** is a block diagram of a computer for acting as a gaming system for implementing embodiments of wagering games in accordance with this disclosure.

[0022] FIG. **14** illustrates an embodiment of data flows between various applications/services for supporting the game, feature or utility of the present disclosure for mobile/interactive gaming.

[0023] While the invention is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail herein. It should be understood, however, that the invention is not intended to be limited to the particular forms disclosed. Rather, the invention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

[0024] While this invention is susceptible of embodiment in many different forms, there is shown in the drawings, and will herein be described in detail, preferred embodiments of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to the embodiments illustrated. For purposes of the present detailed description, the singular includes the plural and vice versa (unless specifically disclaimed); the words “and” and “or” shall be both conjunctive and disjunctive; the word “all” means “any and all”; the word “any” means “any and all”; and the word “including” means “including without limitation.”

[0025] For purposes of the present detailed description, the terms “wagering game,” “casino wagering game,” “gambling,” “slot game,” “casino game,” and the like include games in which a player places at risk a sum of money or other representation of value, whether or not redeemable for cash, on an event with an uncertain outcome, including without limitation those having some element of skill. In some embodiments, the wagering game involves wagers of real money, as found with typical land-based or online casino games. In other embodiments, the wagering game additionally, or alternatively, involves wagers of non-cash values, such as virtual currency, and therefore may be considered a social or casual game, such as would be typically available on a social networking web site, other web sites, across computer networks, or applications on mobile devices (e.g., phones, tablets, etc.). When provided in a social or casual game format, the wagering game may closely resemble a traditional casino game, or it may take another form that more closely resembles other types of social/casual games.

[0026] FIG. **1** is a diagram of an example gaming system **100** with a light-tracking chip tray according to one or more embodiments of the present disclosure. The gaming system **100** includes an overhead view of a gaming table **101** and a light-tracking, chip tray (e.g., chip tray **130**). The chip tray **130** can hold gaming tokens, such as gaming chips (“chips **131**”), tiles, etc., which a

dealer can use to exchange a player's money for physical gaming tokens. The chips **131** rest within one or more vertical, semi-cylindrical slots or columns (e.g., columns **132**) of the chip tray **130**. Each of the columns **132** has a separate sensor array **145**. Each sensor array **145** has a plurality of light sensors **155** positioned inside the columns **132** such that when any one of the chips **131** is placed inside the column **132**, the edge of the chip physically covers one of the light sensors **155**. When one of the chips **131** covers one of the light sensors **155**, it prevents ambient light **121** from reaching the blocked one of the light sensors **155**. The ambient light **121** comes from one or more light sources **120** from the gaming environment at, or around, around the gaming table **101**. Examples of light sources **120** from the gaming environment may include overhead lighting, effects lighting, signage, attract lighting, emotive lighting, game display lighting, light projected from a projector **103**, etc. The light sensors **155** are photoelectric devices that convert light energy (photons) whether visible or infra-red light into an electrical (electrons) signal. For instance, in one embodiment, the light sensors **155** are light dependent resistors (LDRs). In another embodiment, the light sensors **155** are photodiode-based proximity sensors.

[0027] The arrangement of the sensors arrays **145** are shown in the example of FIG. **1** to be aligned vertically along a column **132** and are incorporated with (e.g., affixed to, embedded into, etc.) the physical structure of the chip tray **130**, such as at the bottom of the columns **132** (e.g., see FIG. **4A**, **4B**, and **4C** for more details).

[0028] A controller (e.g., tracking controller **204**) is configured to electronically track lighting levels of each sensor in the sensor arrays **145** to determine which sensors are blocked by chips **131**.

[0029] In some embodiments, the gaming system **100** includes a camera **102** and an indicator device that presents data about the chips, such the projector **103**, a display, speakers, etc. In some embodiments, the camera **102** captures a stream of images of a gaming area, such as an area encompassing a top surface **104** of the gaming table **101**. The projector **103** can also project images of gaming content toward the surface **104** relative to objects in the gaming area. In some instances, the projector **103** is configured to project images of gaming content relevant to some elements of a wagering game that are common, or related, to any or all participants (e.g., the projector **103** projects gaming content at a communal presentation area **114**). In some instances, the camera **102** is positioned above the surface **104** and to the left of a first player area **105**. The camera **102** has a lens that is pointed at the gaming table **101** in a way that views portions of the surface **104** relevant to game play (such as the chip tray **130**) and that views game participants (e.g., players, dealer, back-betting patrons, etc.) positioned around the gaming table **101** (at the different player areas **105**, **106**, **107**, **108**, **109**, and **110**). In some embodiments, the camera **102** also views the chip tray **130**. The projector **103** is also positioned above the gaming table **101**, and also to the left of the first player area **105**. In some instances, the tracking controller **204** also analyzes images taken by the camera **102** (or other cameras positioned at the gaming table **101**). For example, the tracking controller **204** can perform feature set extraction, object classification, etc. via use of a machine-learning model (e.g., a neural network model). The machine-learning model is trained to identify chips as objects and classify the chips according to denomination value based on observation of the color patterns on the edges of the chips.

[0030] In some embodiments, the tracking controller **204** is also configured to automatically detect physical objects in a gaming environment as points of interest based on electronic analysis of an image performed by one or more additional neural network models. For example, the gaming system **100** can detect one or more points of interest by detecting, via a neural network model, physical features of the image that appear at the surface **104**. For example, the tracking controller **204** is configured to monitor the gaming area (e.g., physical objects within the gaming area), and determine a relationship between one or more of the objects. The tracking controller **204** can further receive and analyze collected sensor data (e.g., receives and analyzes the captured image data from the camera **102**) to detect and monitor physical objects. The tracking controller **204** can establish data structures relating to various physical objects detected in the image data. For

example, the tracking controller **204** can apply one or more image neural network models during image analysis that are trained to detect aspects of physical objects. In at least some embodiments, each model applied by the tracking controller **204** may be configured to identify a particular aspect of the image data and provide different outputs for any physical object identified such that the tracking controller **204** may aggregate the outputs of the neural network models together to identify physical objects as described herein. The tracking controller **204** may generate data objects for each physical object identified within the captured image data. The data objects may include identifiers that uniquely identify the physical objects such that the data stored within the data objects is tied to the physical objects. The tracking controller **204** can further store data in a database, such as database system **208** in FIG. 2.

[0031] In some embodiments, the tracking controller **204** is configured to detect bank-change events, or in other words, events that occur in the gaming environment that would affect a change to the overall value of the bank of chips **131** within the chip tray **130**, such as buy-ins, won bets, and pay-outs. For example, the tracking controller **204** identifies betting circles (e.g., main betting circles **105A**, **106A**, **107A**, **108A**, **109A**, and **110A** (“**105A-110A**”) and secondary betting circles **105B**, **106B**, **107B**, **108B**, **109B**, and **110B** (“**105B-110B**”). The tracking controller **204** also detects placement of gaming chips (e.g., as stacks) within the betting circles during betting on a wagering game conducted at the gaming table **101**. The tracking controller **204** can further determine the values of chip stacks within the betting circles. The tracking controller **204** determines, based on the values of the chip stacks, amounts by which the bank is expected to change based on collection of losing bets and/or payouts required for winning bets. The tracking controller **204** can compare the expected amounts to actual changes to the chips **131** in the chip tray **130**. Based on the comparison, the tracking controller **204**, for instance, determines whether there are any errors in placement of chips of one denomination value into a column for a different denomination value. The tracking controller **204** can further generate warnings (e.g. of the errors of placement of chips in the wrong column) and/or generate reports that tracks the accuracy of a dealer's handling of the chips into and out of the bank.

[0032] Some objects may be included at the gaming table **101**, such as cards, a card shoe, dice, etc. but are not shown in FIG. 1 for simplicity of description.

[0033] FIG. 2 is an architectural diagram of an exemplary gaming system **200** having a chip tray, such as chip tray **130**, according to one or more embodiments of the present disclosure. In the example embodiment, the gaming system **200** includes a game controller **202**, the tracking controller **204**, the chip tray **130**, and a tracking database system **208**. The chip tray **130** includes a light-sensor system **206** configured to track a level of ambient light at various sensor positions with the chip tray **130**. In some embodiments, the light-sensor system **206** includes a plurality of light sensors **155** arranged in sensor arrays **145** and affixed to the physical structure of the chip tray **130**. In other embodiments, the gaming system **200** may include additional, fewer, or alternative components, including those described elsewhere herein. Furthermore, in some embodiments, the chip tray **130** includes one or more input devices (“input device(s) **207**”) as well as one or more output devices (“output device(s) **209**”). For example, the input devices may include controls (e.g., buttons, a mouse, wireless control devices, etc.) that an operator (e.g., a dealer, a pit boss, etc.) can use to program information, such as column names, denomination values, warning messages, default amounts, etc. . . . In some instances, the output device(s) **209** may include one or more displays, digital read-outs, speakers, etc. (e.g., see electronic displays **475** and/or **575** on the chip tray **130** as described in FIG. 4 and FIG. 5).

[0034] The gaming area **201** is an environment in which one or more casino wagering games are provided. In the example embodiment, the gaming area **201** is a casino gaming table and the area surrounding the table (e.g., as in FIG. 1). In other embodiments, other suitable gaming areas **201** may be monitored by the gaming system **200**. For example, the gaming area **201** may include one or more floor-standing electronic gaming machines. In another example, multiple gaming tables

may be monitored by the gaming system **200**. Although the description herein may reference a gaming area (such as gaming area **201**) to be a single gaming table and the area surrounding the gaming table, it is to be understood that other gaming areas **201** may be used with the gaming system **200** by employing the same, similar, and/or adapted details as described herein.

[0035] The game controller **202** is configured to facilitate, monitor, manage, and/or control gameplay of the one or more games at the gaming area **201**. More specifically, the game controller **202** is communicatively coupled to at least one or more of the tracking controller **204**, the sensor system **206**, the tracking database system **208**, a gaming device **210**, an external interface **212**, and/or a server system **214** to receive, generate, and transmit data relating to the games, the players, the chips **131** and/or the gaming area **201**. The game controller **202** may include one or more processors, memory devices, and communication devices to perform the functionality described herein. More specifically, the memory devices store computer-readable instructions that, when executed by the processors, cause the game controller **202** to function as described herein, including communicating with the devices of the gaming system **200** via the communication device(s).

[0036] The game controller **202** may be physically located at the gaming area **201** as shown in FIG. 2 or remotely located from the gaming area **201**. In certain embodiments, the game controller **202** may be a distributed computing system. That is, several devices may operate together to provide the functionality of the game controller **202**. In such embodiments, at least some of the devices (or their functionality) described in FIG. 2 may be incorporated within the distributed game controller **202**.

[0037] The gaming device **210** is configured to facilitate one or more aspects of a game. For example, for card-based games, the gaming device **210** may be a card shuffler, shoe, or other card-handling device. The external interface **212** is a device that presents information to a player, dealer, or other user and may accept user input to be provided to the game controller **202**. In some embodiments, the external interface **212** may be a remote computing device in communication with the game controller **202**, such as a player's mobile device. In other examples, the gaming device **210** and/or external interface **212** includes one or more projectors. The server system **214** is configured to provide one or more backend services and/or gameplay services to the game controller **202**. For example, the server system **214** may include accounting services to monitor wagers, payouts, and jackpots for the gaming area **201**. In another example, the server system **214** is configured to control gameplay by sending gameplay instructions or outcomes to the game controller **202**. It is to be understood that the devices described above in communication with the game controller **202** are for exemplary purposes only, and that additional, fewer, or alternative devices may communicate with the game controller **202**, including those described elsewhere herein.

[0038] In the example embodiment, the tracking controller **204** is in communication with the game controller **202**. In other embodiments, the tracking controller **204** is integrated with the game controller **202** such that the game controller **202** provides the functionality of the tracking controller **204** as described herein. Like the game controller **202**, the tracking controller **204** may be a single device or a distributed computing system. In one example, the tracking controller **204** may be at least partially located remotely from the gaming area **201**. That is, the tracking controller **204** may receive data from one or more devices located at the gaming area **201** (e.g., the game controller **202** and/or the sensor system **206**), analyze the received data, and/or transmit data back based on the analysis.

[0039] In the example embodiment, the tracking controller **204**, similar to the example game controller **202**, includes one or more processors, a memory device, and at least one communication device. The memory device is configured to store computer-executable instructions that, when executed by the processor(s), cause the tracking controller **204** to perform the functionality of the tracking controller **204** described herein. The communication device is configured to communicate

with external devices and systems using any suitable communication protocols to enable the tracking controller **204** to interact with the external devices and integrates the functionality of the tracking controller **204** with the functionality of the external devices. The tracking controller **204** may include several communication devices to facilitate communication with a variety of external devices using different communication protocols.

[0040] The tracking controller **204** is configured to monitor at least one or more aspects of the gaming area **201**. In the example embodiment, the tracking controller **204** is configured to monitor physical objects within the area **201**, and determine a relationship between one or more of the objects. Some objects may include gaming tokens. The tokens may be any physical object (or set of physical objects) used to place wagers. As used herein, the term “stack” refers to one or more gaming tokens physically grouped together. For circular tokens typically found in casino gaming environments (e.g., gaming chips), these may be grouped together into a vertical stack (e.g., when placed onto a betting spot) or grouped vertically within a column of a chip tray, such as any of the columns **132** of the chip tray **130**. In another example in which the tokens are monetary bills and coins, a group of bills and coins may be considered a “stack” based on the physical contact of the group with each other and other factors as described herein.

[0041] In the example embodiment, the tracking controller **204** is communicatively coupled to the sensor system **206** to monitor the gaming area **201**. More specifically, the sensor system **206** includes one or more sensors configured to collect sensor data associated with the gaming area **201** (e.g., light sensors **155** of the chip tray **130**), and the tracking controller **204** receives and analyzes the collected sensor data to detect and monitor physical objects (e.g., chips **131**). The sensor system **206** may include any suitable number, type, and/or configuration of sensors to provide sensor data to the game controller **202**, the tracking controller **204**, and/or another device that may benefit from the sensor data.

[0042] In another embodiment, the sensor system **206** includes at least one image sensor that is oriented to capture image data of physical objects in the gaming area **201**. In one example, the sensor system **206** may include a single image sensor that monitors the gaming area **201**. In another example, the sensor system **206** includes a plurality of image sensors that monitor subdivisions of the gaming area **201**. The image sensor may be part of a camera unit of the sensor system **206** or a three-dimensional (3D) camera unit in which the image sensor, in combination with other image sensors and/or other types of sensors, may collect depth data related to the image data, which may be used to distinguish between objects within the image data. The image data is transmitted to the tracking controller **204** for analysis as described herein. In some embodiments, the image sensor is configured to transmit the image data with limited image processing or analysis such that the tracking controller **204** and/or another device receiving the image data performs the image processing and analysis. In other embodiments, the image sensor may perform at least some preliminary image processing and/or analysis prior to transmitting the image data. In such embodiments, the image sensor may be considered an extension of the tracking controller **204**, and as such, functionality described herein related to image processing and analysis that is performed by the tracking controller **204** may be performed by the image sensor (or a dedicated computing device of the image sensor). In certain embodiments, the sensor system **206** may include, in addition to or instead of the image sensor, one or more sensors configured to detect objects, such as light sensors, time-of-flight sensors, radar sensors (e.g., LIDAR), thermographic sensors, and the like.

[0043] The tracking controller **204** is configured to establish data structures relating to various physical objects detected in the collected data from the sensor(s). For example, the tracking controller **204** applies one or more image neural network models during image analysis that are trained to detect aspects of physical objects. Neural network models are analysis tools that classify “raw” or unclassified input data without requiring user input. That is, in the case of the raw image data captured by the image sensor, the neural network models may be used to translate patterns

within the image data to data object representations of, for example, tokens, faces, hands, etc., thereby facilitating data storage and analysis of objects detected in the image data as described herein.

[0044] At a simplified level, neural network models are a set of node functions that have a respective weight applied to each function. The node functions and the respective weights are configured to receive some form of raw input data (e.g., image data), establish patterns within the raw input data, and generate outputs based on the established patterns. The weights are applied to the node functions to facilitate refinement of the model to recognize certain patterns (i.e., increased weight is given to node functions resulting in correct outputs), and/or to adapt to new patterns. For example, a neural network model may be configured to receive input data, detect patterns in the image data representing human body parts, perform image segmentation, and generate an output that classifies one or more portions of the image data as representative of segments of a player's body parts (e.g., a box having coordinates relative to the image data that encapsulates a face, an arm, a hand, etc. and classifies the encapsulated area as a "human," "face," "arm," "hand," etc.).

[0045] For instance, to train a neural network to identify the most relevant guesses for identifying a human body part, for example, a predetermined dataset of raw image data including image data of human body parts, and with known outputs, is provided to the neural network. As each node function is applied to the raw input of a known output, an error correction analysis is performed such that node functions that result in outputs near or matching the known output may be given an increased weight while node functions having a significant error may be given a decreased weight. In the example of identifying a human face, node functions that consistently recognize image patterns of facial features (e.g., nose, eyes, mouth, etc.) may be given additional weight. Similarly, in the example of identifying a human hand, node functions that consistently recognize image patterns of hand features (e.g., wrist, fingers, palm, etc.) may be given additional weight. The outputs of the node functions (including the respective weights) are then evaluated in combination to provide an output such as a data structure representing a human face. Training may be repeated to further refine the pattern-recognition of the model, and the model may still be refined during deployment (i.e., raw input without a known data output).

[0046] At least some of the neural network models applied by the tracking controller **204** may be deep neural network (DNN) models. DNN models include at least three layers of node functions linked together to break the complexity of image analysis into a series of steps of increasing abstraction from the original image data. For example, for a DNN model trained to detect human faces from an image, a first layer may be trained to identify groups of pixels that represent the boundary of facial features, a second layer may be trained to identify the facial features as a whole based on the identified boundaries, and a third layer may be trained to determine whether or not the identified facial features form a face and distinguish the face from other faces. The multi-layered nature of the DNN models may facilitate more targeted weights, a reduced number of node functions, and/or pipeline processing of the image data (e.g., for a three-layered DNN model, each stage of the model may process three frames of image data in parallel).

[0047] In at least some embodiments, each model applied by the tracking controller **204** may be configured to identify a particular aspect of the image data and provide different outputs such that the tracking controller **204** may aggregate the outputs of the neural network models together to identify physical objects as described herein. For example, one model may be trained to identify human faces, while another model may be trained to identify the bodies of players. In such an example, the tracking controller **204** may link together a face of a player to a body of the player by analyzing the outputs of the two models. In other embodiments, a single DNN model may be applied to perform the functionality of several models.

[0048] As described in further detail below, the tracking controller **204** may generate data objects for each physical object identified within the captured image data by the DNN models. The data objects are data structures that are generated to link together data associated with corresponding

physical objects. For example, the outputs of several DNN models associated with a player may be linked together as part of a player data object.

[0049] It is to be understood that the underlying data storage of the data objects may vary in accordance with the computing environment of the memory device or devices that store the data object. That is, factors such as programming language and file system may vary the where and/or how the data object is stored (e.g., via a single block allocation of data storage, via distributed storage with pointers linking the data together, etc.). In addition, some data objects may be stored across several different memory devices or databases.

[0050] In some embodiments, the player data objects include a player identifier, and data objects of other physical objects include other identifiers. The identifiers uniquely identify the physical objects such that the data stored within the data objects is tied to the physical objects. In some embodiments, the identifiers may be incorporated into other systems or subsystems. For example, a player account system may store player identifiers as part of player accounts, which may be used to provide benefits, rewards, and the like to players. In certain embodiments, the identifiers may be provided to the tracking controller **204** by other systems that may have already generated the identifiers.

[0051] In at least some embodiments, the data objects and identifiers may be stored by the tracking database system **208**. The tracking database system **208** includes one or more data storage devices (e.g., one or more databases) that store data from at least the tracking controller **204** in a structured, addressable manner. That is, the tracking database system **208** stores data according to one or more linked metadata fields that identify the type of data stored and can be used to group stored data together across several metadata fields. The stored data is addressable such that stored data within the tracking database system **208** may be tracked after initial storage for retrieval, deletion, and/or subsequent data manipulation (e.g., editing or moving the data). The tracking database system **208** may be formatted according to one or more suitable file system structures (e.g., FAT, exFAT, ext4, NTFS, etc.).

[0052] The tracking database system **208** may be a distributed system (i.e., the data storage devices are distributed to a plurality of computing devices) or a single device system. In certain embodiments, the tracking database system **208** may be integrated with one or more computing devices configured to provide other functionality to the gaming system **200** and/or other gaming systems. For example, the tracking database system **208** may be integrated with the tracking controller **204** or the server system **214**.

[0053] In the example embodiment, the tracking database system **208** is configured to facilitate a lookup function on the stored data for the tracking controller **204**. The lookup function compares input data provided by the tracking controller **204** to the data stored within the tracking database system **208** to identify any “matching” data. It is to be understood that “matching” within the context of the lookup function may refer to the input data being the same, substantially similar, or linked to stored data in the tracking database system **208**. For example, if the input data is an image of a player's face, the lookup function may be performed to compare the input data to a set of stored images of historical players to determine whether or not the player captured in the input data is a returning player. In this example, one or more image comparison techniques may be used to identify any “matching” image stored by the tracking database system **208**. For example, key visual markers for distinguishing the player may be extracted from the input data and compared to similar key visual markers of the stored data. If the same or substantially similar visual markers are found within the tracking database system **208**, the matching stored image may be retrieved. In addition to or instead of the matching image, other data linked to the matching stored image may be retrieved during the lookup function, such as a player account number, the player's name, etc. In at least some embodiments, the tracking database system **208** includes at least one computing device that is configured to perform the lookup function. In other embodiments, the lookup function is performed by a device in communication with the tracking database system **208** (e.g., the tracking

controller **204**) or a device in which the tracking database system **208** is integrated within.

[0054] FIG. **3** is a diagram of an exemplary sensor system (e.g., sensor system **206**) for the chip tray **130** according to one or more embodiments of the present disclosure. In the example illustrated in FIG. **3**, the sensor system **206** includes at least one of a plurality of column modules **301** for each one of the columns **132** of the chip tray **130**. In some embodiments, each one of the column modules **301** is equivalent to each one of the sensor arrays **145** positioned in each of the columns **132**. In one embodiment. The sensor system **206** utilizes LDRs **355** as examples of the light sensors **155** described previously. Each of the LDRs **355** are connected to a resistor **356** in a voltage divider circuit, the output of which is connected to a comparator **357**. The combination of the LDR **355**, the resistor **356**, and the comparator **357** may be referred to more succinctly as a sensor unit **350**. Each of the comparators **357** connect the voltage divider circuits to a microcontroller **358** as shown in FIG. **3**. The microcontroller **358** is connected to a communication bus **360**. Each of the column modules **301** are connected to each other via the communication bus **360**. Each one of the comparators **357** compares an output from its respective one of the LDRs **355** to a calibrated threshold value (measured by a threshold sensor **359**). The threshold sensor **359** may be equivalent to a sensor unit **350** (e.g., it includes an LDR **355**, a resistor **356**, and a comparator **357**). The threshold sensor **359** is calibrated to the level of environmental lighting in the gaming environment. In some instances, the threshold sensor **359** is not positioned in the column **132** and is not covered by one of the chips **131**. In another embodiment, the threshold sensor **359** is the sensor that is at the highest portion (the top vertical portion) of a sensor array **145**, such as sensor **556** as shown in FIG. **5**. For instance, as described in more detail in FIG. **4C**, a column **132** has a downward slope (e.g., see column **132** which has slight angle of decline **414**). In other words, when the bottom portion **416** of the chip tray **130** is resting against a flat plane on the gaming table **101**, the columns **132** slope downward so that the chips **131** slide downward within their respective column **132** due to gravity. Consequently, the highest sensor in the array **145** would be the last to be covered up by a chip **131** as all other chips **131** in the given column would slide downward within the column **132**. Thus, if the threshold sensor **359** is at the top of the array as shown in FIG. **5**, then it would be the last sensor in the array to be covered up, and thus can function as the threshold array **359** until it is covered up. If the amount of light sensed from the threshold sensor **359** is less than that of any other threshold sensor **349** in another column, then the column **132** may be considered full. In other embodiments, the threshold sensor **359** is not covered (e.g., it is outside of the column **132**), and thus functions as a baseline measurement for the level of ambient light at all times. In some embodiments, the sensor system **206** utilizes a single threshold sensor **359** as a baseline, ambient-light reference for the entire chip tray **130**.

[0055] The components of the sensor system **206** (e.g., the LDRs **355**, the comparators **357**, the resistors **356**, the microcontroller **358**, etc.) may be surface mounted devices (SMDs) placed on a printed circuit board (PCB) connected with etched circuit lines (e.g., see printed-circuit board **460** in FIG. **4**). The tracking controller **204** receives data from the light-sensor system **206** for each of the column modules **301**. Thus, the tracking controller **204** calculates the number of chips **131** in each column **132**. In some embodiments, the data is sent via an inter-integrated circuit (I2C) communication protocol. The tracking controller **204** can access each module using the bus **360** by addressing each column module **301** individually.

[0056] FIG. **4A**, **4B**, and **4C** are diagrams that illustrate different perspective views of a column of a chip tray according to one or more embodiments of the present disclosure FIG. **4A** illustrates a portion of a single one of the columns **132**. Different portions of the same one of the columns **132** are illustrated in FIG. **4B** and **4C** from different viewing angles. In FIG. **4A**, a display is shown **475** that indicates a value **476** of the chips **131** within the given column **132**. As illustrated in FIG. **4B**, the given column **132** may be designated for chips **131** having a denomination of “twenty-five” betting units (e.g., a \$25 chip). Consequently, the tracking controller **204** detects, via the sensor system **206**, the number of chips **131** within the given column **132** (designated for \$25 chips). The

tracking controller **204**, for instance, detects the presence of five chips in the column **132**. The tracking controller **204** multiplies the five chips by twenty-five dollars each, which computes to one-hundred twenty five dollars (\$125). As a result, the display **475** presents (e.g., digitally illustrates) the value **476** of “125.” In some embodiments, the chip tray **130** also includes a display (not shown) that illustrates a total amount of chips in the chip tray **130** at any given time.

[0057] The arrangement of the sensors arrays **145** can be aligned vertically along a column **132** and incorporated with the physical structure of the chip tray **130**, such as at the bottom of the columns **132**. For example, openings (e.g., holes **448**) are formed through the material of the chip tray **130** at the bottom of each of the columns **132** such that the sensors **155** are positioned under the holes **448**. In some embodiments, the sensors in the sensor array **145** are positioned a given distance apart from each other. In some embodiments, the given distance from each other corresponds to the thickness (i.e., the height) of a standard chip (e.g., approximately 3.3 mm). Consequently, in some embodiments, the chip tray **130** is configured so that a single one of light sensors **155** corresponds to a single one of the given chips **131** (i.e., in a 1:1 ratio). Furthermore, because (according to some embodiments) the spacing of any given one of the sensors **155** corresponds to the thickness of a single one of the chips **131**, then the spacing **440** between the holes **448** (through which the sensors **144** are exposed) are less than the thickness of a standard chip.

[0058] A transparent material is positioned over the holes **448**, such as a window or a lens (e.g., windows **438**), to cover and protect the sensor **155** from dust or debris. The material of the windows **438** is transparent to permit the ambient light **121** to shine through it to reach the one of the sensors **155** beneath. In some instances, the window **438** is shaped as a lens to focus or amplify the ambient light shining through the window **438** onto one of the respective sensors **155**. Furthermore, the transparent barrier of the window **438** protects the sensor **155** from being directly exposed to dirt or debris.

[0059] In some embodiments, the light sensors **155** are organized into a vertical array. In one embodiment, the light sensors **155** are uniformly spaced and mounted underneath the chip tray **430** along an underside edge **412** of the column **432**.

[0060] In some embodiments, the light sensors **155** are embedded into the material of the chip tray **130**. In some embodiments, a line of the holes **448** are drilled through the material of an interior curved wall **439** of the chip tray **130** along a bottom of the column **132**. In some embodiments, a given light sensor **155** is positioned inside each drilled hole approximately 1 mm or less in distance from the interior curved wall **439** of the chip column **132** (upon which rests the edge **480** of any of the given chips **131**).

[0061] As referred to briefly in the description related to FIG. 3, in some embodiments, the columns **132** are sloped downward at the slight angle of decline **414** (e.g., at approximately a 5 degree angle of decline). In other words, when the bottom portion **416** of the chip tray **130** is resting against a flat plane on the gaming table **101**, the columns **132** slope downward so that the chips **131** slide downward (due to gravity). Thus, the array **145**, from the perspective of the chip tray **130**, appears to be sloping downward. However, the faces of the light sensors **155** are aligned to be approximately perpendicular to the bottom portion of the interior curved wall **439**. In other words, the sensing faces of the light sensors **155** are positioned to be approximately perpendicular to the chip edges **480**, even though from the perspective of the chip tray **130**, the light sensors **155** themselves are at the slight angle of decline **414**.

[0062] FIG. 5 is a diagram of an exemplary chip tray (e.g., chip tray **130**) according to one or more embodiments of the present disclosure. In FIG. 5, each individual sensor **155** from sensor arrays **145** detects a level of ambient light. The controller **204** (e.g. see FIG. 2) analyzes data received from each of the sensors **155**, and tracks data for each column **132** individually (e.g., to determine a level of ambient light reaching each sensor **155** in a column **132**), and collectively for all columns **132** of the chip tray **130**.

[0063] Each column **132** has a unique identifier (e.g. column A, column B, column C, and so

forth). In some embodiments, the column names may correspond to standard chip denomination values (e.g., “1,” “5,” “10,” “25,” “100,” “500,” etc.). In some embodiments, the chip tray **130** may include markings to indicate the column names (e.g., printed markings, digital illustrations, etc.). In some instances, the chip tray **130** includes electronic displays **575**, or other such devices, that graphically or digitally illustrates a value for any given column. In some instances, the chip tray **130** is customizable. For example, the chip tray **130** may include controls that an operator (e.g., a dealer, a pit boss, etc.) can use to program column names to specify denomination values (e.g., via the electronic displays **575**). The displays **575** may be interchangeable in function with displays **475** (see FIG. 4A). In other words, displays **475** may be utilized to designate a column name whereas displays **575** may be utilized to designate a current total chip value for any given column **132**.

[0064] In some embodiments, given that each sensor array **145** is affixed relative to its given column **132**, each sensor **155** can be associated with a location identifier **518** for any given chip within any given column. The location identifier **518** may be stored in a data map (e.g., in database system **208**) that identifies a physical location of each sensor **155** within any given sensor array **145** within the chip tray **130** and maps the physical location to a coordinate (or group of coordinates) on a two-dimensional overlay grid (e.g., coordinates grid). Thus every given sensor **155** in an array **145** can be associated with a single one of the chips **131** for any chip stack in the chip tray **130**. Given the known height of a standard chip, the data map is constructed by mapping the height **515** upward from the bottom of a column to the top of a column. For example, given the known height **515**, the map thus determines which sensors from the sensor array **145** are mapped to given coordinates (e.g., to a specific chip **131**). For instance, light sensors **155** may be designated with coordinates indicating the location identifier **518** of “(A, 1),” thus specifying the column identifier (e.g., column “A”) and the height identifier (e.g., “1”). In the example shown in FIG. 5, the number of light sensors **155** corresponds to one location identifier **518** in a 1-to-1 ratio (i.e., one light sensor **155** per location identifier **518**). However, in other embodiments, more than one light sensor **155** can correspond to a single location identifier **518**. For instance, the arrays **145** may be constructed to have multiple light sensors within the space **519** of one height identifier. For example, the light sensors **155** may be smaller than a thickness of a standard chip (i.e., multiple light sensors for the height of one chip). Multiple sensors per chip can provide more data to the tracking controller **204**. Furthermore, multiple sensors per chip can provide redundancy should one of the light sensors **155** fail within the array **145** during operation.

[0065] FIG. 6 is a flow diagram of an example method flow (flow **600**) according to one or more embodiments of the present disclosure.

[0066] In FIG. 6, flow **600** begins at processing block **602** where a processor (e.g., tracking controller **204**), detects, via an array of light sensors positioned within a chip tray column, a level of ambient light at each sensor location with the array. In some instances, the tracking controller **204** detects the level of ambient light received at each sensor location by comparing the level of light received at any given sensor to that of a threshold sensor, which detects the level of ambient light at the chip tray **130**. In some embodiments, the chip tray **130** includes one threshold sensor per each column **132**. In other embodiments, the chip tray **130** includes less than a single threshold sensor for each column **132**, for example, a single threshold sensor for an entire chip tray **130**, or a limited set of threshold sensors placed at different locations on the chip tray **130**. For example, the chip tray **130** may include threshold sensors at each corner of the chip tray **130** and the tracking controller **204** may utilize an average level of ambient light received at each of the threshold sensors as a baseline, or threshold value of the amount of ambient light in the gaming environment at or around the gaming table **101** that shines onto the chip tray **130** without being blocked by a chip. Thus, the tracking controller **204** can compare any detected level of light at any given position of one of the sensors **155** to the threshold value (detected via the threshold sensor(s)) to determine whether the given sensor **155** (at the given position) is being blocked by a chip (thus indicating the

presence of a chip at the given location of the given sensor **155**).

[0067] Referring back to FIG. **6**, the flow **600** continues at processing block **604**, where the tracking controller **204** determines, in response to detecting the level of ambient light at each sensor location, a number of gaming chips placed in the chip column. For example, as shown in FIG. **4C**, each of the chips **131** covers a respective one of the sensors **155** in the array **145**. Thus, the tracking controller **204** detects that the level of ambient light at each of those blocked sensors is less than that of the value detected by a threshold sensor. For example, if the tracking controller **204** detects a five percent (5%) or more difference in light levels at any given sensor **155** from that of the threshold value, then the tracking controller **204** determines that at least one of the gaming chips **131** is present at the sensor location (i.e., the tracking controller **204** determines that a chip is blocking the ambient light at the sensor location).

[0068] Referring back to FIG. **6**, the flow **600** continues at processing block **606** where the tracking controller **204** computes, in response to determining the number of gaming chips, a chip-column value. In some embodiments, the tracking controller **204** determines the chip-column value based on at least the number of chips in any given column. For example, as shown in FIG. **5**, one of the columns **132** (e.g., the column designated as column “B” in FIG. **5**), has five chips **131**. Column “B” is designated for chips having the denomination value of “twenty-five” (e.g., \$25 chips). Thus, the dealer only stacks \$25 chips into that column. As a consequence, the tracking controller **204** multiplies the value of “5” (representing the five detected chips **131**) by the value of “25” (representing the designated denomination value for the given column), resulting in an amount of “125.”

[0069] In some instances, the tracking controller **204** detects a color pattern for chips placed in a given column **132**. For instance, a dealer may track the input and output of chips **131** from the chip tray **130** by placing chips of only a certain denomination value into a designated subset of the columns **132**. For example, multiple ones of the columns **132** may be designated to hold lower chip values (e.g., multiple ones of the columns **132** may be designated to hold chips having a value of “1,” “5,” and “10” (e.g., \$1, \$5, \$10); whereas each higher denomination (e.g., “25,” “50,” “100,” “500,” etc.), may be designated to only one of the columns **132** respectively. The tracking controller **204**, thus, can optionally detect, via analysis of images of the chip tray **130**, whether any of the columns **132** has a chip with an edge-color pattern that does not match that of others in the designated column. For example, the tracking controller **204** can determine that a certain percentage of the chips **131** in any given column have the same given edge-color pattern. In some embodiment, the tracking controller **204** can determine (e.g., deduce) that the edge-color pattern that appears with the most frequency amongst the total number of chips in a given column is the default edge-color pattern for the designated denomination value for that given column. For example, in a column with at least three chips, if two or more possess the same edge-color pattern, the tracking controller **204** can deduce that the chip-denomination value designated for the particular column equates to the chip-denomination value associated with that most-frequently-occurring edge-color pattern. In another embodiments, the tracking controller **204** deduces the chip-denomination value associated with the column by determining the value associated with the edge-color pattern for a majority (or at least a relative majority) of the chips (i.e., the number or part of the total chips having the particular edge-color being greater than any other number or part). The tracking controller **204** can then notify an operator (e.g., the dealer), that the mismatched chip is misplaced within an incorrect one of the columns **132**, and instead should be placed into a correct one of the columns **132** designated for the specific denomination value. In some embodiments, the tracking controller **204** utilizes the camera **102** (see FIG. **1**), to capture images of the chip tray **130**. The tracking controller **204** can analyze the images, using a machine learning model (e.g. a neural network model), trained to detect the chip colors and/or edge patterns that distinguish the different chip denomination values. In some instances, the tracking controller **204** determines, via the machine model analysis, whether a chip is an incorrect column (e.g., detects whether a \$25 chip is

in a column designated for \$5 chips). In some instances, the tracking controller **204** can detect, via electronic image analysis, which columns are designated to each chip denomination value. For example, in some instances, some dealers may switch up chip columns for different chip denomination values throughout a gaming session depending on given gaming conditions. For example, a dealer may have a high-roller or VIP table which requires large betting amounts. The participants may be betting large amounts of chips, and so the dealer may be dealing, during that gaming session, with large denomination values. As a consequence, the dealer may configure the use of the chip tray **130** on the fly, to switch up which columns should be utilized for each given denomination value. However, the dealer may not have the time, during the session, to manually specify column denomination values using manual controls (e.g., using the input device(s) **207** mentioned in FIG. 2). Consequently, in some embodiments, the tracking controller **204** utilizes the machine learning model to automatically detect, in real-time, which columns the dealer is utilizing for given chip denomination values. The tracking controller **204**, thus, can detect column denomination values for a given stack of chips within any given column by detecting a common edge color or pattern amongst the chips already in a column. The tracking controller **204** can further give warnings if a chip pattern is detected (via the image analysis), which is different from other chips in the given column. For example, if the difference in chips is high enough (e.g., above a \$4 difference), then the tracking controller **204** can generate a warning of a potentially misplaced chip or missing chips (generate warning for dealer, pitboss, etc.).

[0070] In some instances, the tracking controller **204** provides real-time warnings (during a gaming session at the gaming table) to indicate potential errors in placement of chips within the stack by the dealer. In some embodiments, the tracking controller **204** evaluates the placement of the chips and provides a report after a dealer session regarding potential errors.

[0071] Referring back to FIG. 6, the flow **600** continues at processing block **608** with electronically indicating the chip stack value. For example, after the tracking controller **204** computes a total monetary value for a chip stack in any given column, the tracking controller **204** can indicate the total monetary value via a display (e.g., see display **475** in FIG. 4). In other embodiments, the tracking controller **204** can indicate the total monetary value via a display associated with an operator or administrator that is not at the gaming table **101** (e.g., via a display associated with a computer or device of a pit boss, an accountant, a security officer, etc.). In other embodiments, the tracking controller **204** can electronically indicate a chip-column value via an audio device, such as via a speaker. In some embodiments, the tracking controller **204** can display a chip-column value in the form of a meter or gauge that displays a level of increase or decrease in value for the chip column.

[0072] FIG. 7 is a perspective view of an embodiment of a gaming table **1200** (which may be configured as the gaming table **101**) for implementing wagering games in accordance with this disclosure. The gaming table **1200** may be a physical article of furniture around which participants in the wagering game may stand or sit and on which the physical objects used for administering and otherwise participating in the wagering game may be supported, positioned, moved, transferred, and otherwise manipulated. For example, the gaming table **1200** may include a gaming surface **1202** (e.g., a table surface) on which the physical objects used in administering the wagering game may be located. The gaming surface **1202** may be, for example, a felt fabric covering a hard surface of the table, and a design, conventionally referred to as a “layout,” specific to the game being administered may be physically printed on the gaming surface **1202**. As another example, the gaming surface **1202** may be a surface of a transparent or translucent material (e.g., glass or plexiglass) onto which a projector, which may be located, for example, above or below the gaming surface **1202**, may illuminate a layout specific to the wagering game being administered. In such an example, the specific layout projected onto the gaming surface **1202** may be changeable, enabling the gaming table **1200** to be used to administer different variations of wagering games within the scope of this disclosure or other wagering games. In either example, the gaming surface

1202 may include, for example, designated areas for player positions; areas in which one or more of player cards, dealer cards, or community cards may be dealt; areas in which wagers may be accepted; areas in which wagers may be grouped into pots; and areas in which rules, pay tables, and other instructions related to the wagering game may be displayed. As a specific, non-limiting example, the gaming surface **1202** may be configured as any table surface described herein.

[0073] In some embodiments, the gaming table **1200** may include a display **1210** separate from the gaming surface **1202**. The display **1210** may be configured to face players, prospective players, and spectators and may display, for example, information randomly selected by a shuffler device and also displayed on a display of the shuffler device; rules; pay tables; real-time game status, such as wagers accepted and cards dealt; historical game information, such as amounts won, amounts wagered, percentage of hands won, and notable hands achieved; the commercial game name, the casino name, advertising and other instructions and information related to the wagering game. The display **1210** may be a physically fixed display, such as an edge lit sign, in some embodiments. In other embodiments, the display **1210** may change automatically in response to a stimulus (e.g., may be an electronic video monitor).

[0074] The gaming table **1200** may include particular machines and apparatuses configured to facilitate the administration of the wagering game. For example, the gaming table **1200** may include one or more card-handling devices **1204A**, **1204B**. The card-handling device **1204A** may be, for example, a shoe from which physical cards **1206** from one or more decks of intermixed playing cards may be withdrawn, one at a time. Such a card-handling device **1204A** may include, for example, a housing in which cards **1206** are located, an opening from which cards **1206** are removed, and a card-presenting mechanism (e.g., a moving weight on a ramp configured to push a stack of cards down the ramp) configured to continually present new cards **1206** for withdrawal from the shoe.

[0075] In some embodiments in which the card-handling device **1204A** is used, the card-handling device **1204A** may include a random number generator **151** and the display **152**, in addition to or rather than such features being included in a shuffler device. In addition to the card-handling device **1204A**, the card-handling device **1204B** may be included. The card-handling device **1204B** may be, for example, a shuffler configured to select information (using a random number generator), to display the selected information on a display of the shuffler, to reorder (either randomly or pseudo-randomly) physical playing cards **1206** from one or more decks of playing cards, and to present randomized cards **1206** for use in the wagering game. Such a card-handling device **1204B** may include, for example, a housing, a shuffling mechanism configured to shuffle cards, and card inputs and outputs (e.g., trays). Shufflers may include card recognition capability that can form a randomly ordered set of cards within the shuffler. The card-handling device **1204** may also be, for example, a combination shuffler and shoe in which the output for the shuffler is a shoe.

[0076] In some embodiments, the card-handling device **1204** may be configured and programmed to administer at least a portion of a wagering game being played utilizing the card-handling device **1204**. For example, the card-handling device **1204** may be programmed and configured to randomize a set of cards and deliver cards individually for use according to game rules and player and or dealer game play elections. More specifically, the card-handling device **1204** may be programmed and configured to, for example, randomize a set of six complete decks of cards including one or more standard 52-card decks of playing cards and, optionally, any specialty cards (e.g., a cut card, bonus cards, wild cards, or other specialty cards). In some embodiments, the card-handling device **1204** may present individual cards, one at a time, for withdrawal from the card-handling device **1204**. In other embodiments, the card-handling device **1204** may present an entire shuffled block of cards that are transferred manually or automatically into a card dispensing shoe **1204**. In some such embodiments, the card-handling device **1204** may accept dealer input, such as, for example, a number of replacement cards for discarded cards, a number of hit cards to add, or a number of partial hands to be completed. In other embodiments, the device may accept a dealer

input from a menu of game options indicating a game selection, which will select programming to cause the card-handling device **1204** to deliver the requisite number of cards to the game according to game rules, player decisions and dealer decisions. In still other embodiments, the card-handling device **1204** may present the complete set of randomized cards for manual or automatic withdrawal from a shuffler and then insertion into a shoe. As specific, nonlimiting examples, the card-handling device **1204** may present a complete set of cards to be manually or automatically transferred into a card dispensing shoe, or may provide a continuous supply of individual cards.

[0077] In another embodiment, the card handling device may be a batch shuffler, such as by randomizing a set of cards using a gripping, lifting, and insertion sequence.

[0078] In some embodiments, the card-handling device **1204** may employ a random number generator device to determine card order, such as, for example, a final card order or an order of insertion of cards into a compartment configured to form a packet of cards. The compartments may be sequentially numbered, and a random number assigned to each compartment number prior to delivery of the first card. In other embodiments, the random number generator may select a location in the stack of cards to separate the stack into two sub-stacks, creating an insertion point within the stack at a random location. The next card may be inserted into the insertion point. In yet other embodiments, the random number generator may randomly select a location in a stack to randomly remove cards by activating an ejector.

[0079] Regardless of whether the random number generator (or generators) is hardware or software, it may be used to implement specific game administrations methods of the present disclosure.

[0080] The card-handling device **1204** may simply be supported on the gaming surface **1202** in some embodiments. In other embodiments, the card-handling device **1204** may be mounted into the gaming table **1202** such that the card-handling device **1204** is not manually removable from the gaming table **1202** without the use of tools. In some embodiments, the deck or decks of playing cards used may be standard, 52-card decks. In other embodiments, the deck or decks used may include cards, such as, for example, jokers, wild cards, bonus cards, etc. The shuffler may also be configured to handle and dispense security cards, such as cut cards.

[0081] In some embodiments, the card-handling device **1204** may include an electronic display **1207** for displaying information related to the wagering game being administered. The electronic display **1207** may display a menu of game options, the name of the game selected, the number of cards per hand to be dispensed, acceptable amounts for other wagers (e.g., maximums and minimums), numbers of cards to be dealt to recipients, locations of particular recipients for particular cards, winning and losing wagers, pay tables, winning hands, losing hands, and payout amounts. In other embodiments, information related to the wagering game may be displayed on another electronic display, such as, for example, the display **1210** described previously.

[0082] The type of card-handling device **1204** employed to administer embodiments of the disclosed wagering game, as well as the type of card deck employed and the number of decks, may be specific to the game to be implemented. Cards used in games of this disclosure may be, for example, standard playing cards from one or more decks, each deck having cards of four suits (clubs, hearts, diamonds, and spades) and of rankings ace, king, queen, jack, and ten through two in descending order. As a more specific example, six, seven, or eight standard decks of such cards may be intermixed. Typically, six or eight decks of 52 standard playing cards each may be intermixed and formed into a set to administer a blackjack or blackjack variant game. After shuffling, the randomized set may be transferred into another portion of the card-handling device **1204B** or another card-handling device **1204A** altogether, such as a mechanized shoe capable of reading card rank and suit.

[0083] The gaming table **1200** may include one or more chip racks **1208** configured to facilitate accepting wagers, transferring lost wagers to the house, and exchanging monetary value for wagering elements **1212** (e.g., chips). For example, the chip rack **1208** (also referred to as a chip

tray herein) may include a series of token support columns, each of which may support tokens of a different type (e.g., color and denomination). In some embodiments, the chip rack **1208** may be configured to automatically present a selected number of chips using a chip-cutting-and-delivery mechanism. In some embodiments, the gaming table **1200** may include a drop box **1214** for money that is accepted in exchange for wagering elements or chips **1212**. The drop box **1214** may be, for example, a secure container (e.g., a safe or lockbox) having a one-way opening into which money may be inserted and a secure, lockable opening from which money may be retrieved. Such drop boxes **1214** are known in the art, and may be incorporated directly into the gaming table **1200** and may, in some embodiments, have a removable container for the retrieval of money in a separate, secure location.

[0084] When administering a wagering game in accordance with embodiments of this disclosure, a dealer **1216** may receive money (e.g., cash) from a player in exchange for wagering elements **1212**. The dealer **1216** may deposit the money in the drop box **1214** and transfer physical wagering elements **1212** to the player. As part of the method of administering the game, the dealer **1216** may accept one or more initial wagers from the player, which may be reflected by the dealer **1216** permitting the player to place one or more wagering elements **1212** or other wagering tokens (e.g., cash) within designated areas on the gaming surface **1202** associated with the various wagers of the wagering game. Once initial wagers have been accepted, the dealer **1216** may remove physical cards **1206** from the card-handling device **1204** (e.g., individual cards, packets of cards, or the complete set of cards) in some embodiments. In other embodiments, the physical cards **1206** may be hand-pitched (i.e., the dealer **1216** may optionally shuffle the cards **1206** to randomize the set and may hand-deal cards **1206** from the randomized set of cards). The dealer **1216** may position cards **1206** within designated areas on the gaming surface **1202**, which may designate the cards **1206** for use as individual player cards, community cards, or dealer cards in accordance with game rules. House rules may require the dealer to accept both main and secondary wagers before card distribution. House rules may alternatively allow the player to place only one wager (i.e., the second wager) during card distribution and after the initial wagers have been placed, or after card distribution but before all cards available for play are revealed.

[0085] In some embodiments, after dealing the cards **1206**, and during play, according to the game rules, any additional wagers (e.g., the play wager) may be accepted, which may be reflected by the dealer **1216** permitting the player to place one or more wagering elements **1212** within the designated area (i.e., area **124**) on the gaming surface **1202** associated with the play wager of the wagering game. The dealer **1216** may perform any additional card dealing according to the game rules. Finally, the dealer **1216** may resolve the wagers, award winning wagers to the players, which may be accomplished by giving wagering elements **1212** from the chip rack **1208** to the players, and transferring losing wagers to the house, which may be accomplished by moving wagering elements **1212** from the player designated wagering areas to the chip rack **1208**.

[0086] FIG. **8** is a perspective view of an individual electronic gaming device **1300** (e.g., an electronic gaming machine (EGM)) configured for implementing wagering games according to this disclosure. The individual electronic gaming device **1300** may include an individual player position **1314** including a player input area **1332** configured to enable a player to interact with the individual electronic gaming device **1300** through various input devices (e.g., buttons, levers, touchscreens). The player input area **1332** may further include a cash- or ticket-in receptor, by which cash or a monetary-valued ticket may be fed, by the player, to the individual electronic gaming device **1300**, which may then detect, in association with game-logic circuitry in the individual electronic gaming device **1300**, the physical item (cash or ticket) associated with the monetary value and then establish a credit balance for the player. In other embodiments, the individual electronic gaming device **1300** detects a signal indicating an electronic wager was made. Wagers may then be received, and covered by the credit balance, upon the player using the player input area **1332** or elsewhere on the machine (such as through a touch screen). Won payouts and pushed or returned

wagers may be reflected in the credit balance at the end of the round, the credit balance being increased to reflect won payouts and pushed or returned wagers and/or decreased to reflect lost wagers.

[0087] The individual electronic gaming device **1300** may further include, in the individual player position **1312**, a ticket-out printer or monetary dispenser through which a payout from the credit balance may be distributed to the player upon receipt of a cashout instruction, input by the player using the player input area **1332**.

[0088] The individual electronic gaming device **1300** may include a gaming screen **1374** configured to display indicia for interacting with the individual electronic gaming device **1300**, such as through processing one or more programs stored in game-logic circuitry providing memory **1340** to implement the rules of game play at the individual electronic gaming device **1300**. Accordingly, in some embodiments, game play may be accommodated without involving physical playing cards, chips or other wagering elements, and live personnel. The action may instead be simulated by a control processor **1350** operably coupled to the memory **1340** and interacting with and controlling the individual electronic gaming device **1300**. For example, the processor may cause the display **1374** to display cards, including virtual player and virtual dealer cards for playing games of the present disclosure.

[0089] Although the individual electronic gaming device **1300** displayed in FIG. **8** has an outline of a traditional gaming cabinet, the individual electronic gaming device **1300** may be implemented in other ways, such as, for example, on a bartop gaming terminal, through client software downloaded to a portable device, such as a smart phone, tablet, or laptop computer. The individual electronic gaming device **1300** may also be a non-portable personal computer (e.g., a desktop or all-in-one computer) or other computing device. In some embodiments, client software is not downloaded but is native to the device or is otherwise delivered with the device when distributed. In such embodiments, the credit balance may be established by receiving payment via credit card or player's account information input into the system by the player. Cashouts of the credit balance may be allotted to a player's account or card.

[0090] A communication device **1360** may be included and operably coupled to the processor **1350** such that information related to operation of the individual electronic gaming device **1300**, information related to the game play, or combinations thereof may be communicated between the individual electronic gaming device **1300** and other devices, such as a server, through a suitable communication medium, such as, for example, wired networks, Wi-Fi networks, and cellular communication networks.

[0091] The gaming screen **1374** may be carried by a generally vertically extending cabinet **1376** of the individual electronic gaming device **1300**. The individual electronic gaming device **1300** may further include banners to communicate rules of game play, instructions, game play advice or hints and the like, such as along a top portion **1378** of the cabinet **1376** of the individual electronic gaming device **1300**. The individual electronic gaming device **1300** may further include additional decorative lights (not shown), and speakers (not shown) for transmitting and optionally receiving sounds during game play.

[0092] Some embodiments may be implemented at locations including a plurality of player stations. Such player stations may include an electronic display screen for display of game information (e.g., cards, wagers, and game instructions) and for accepting wagers and facilitating credit balance adjustments. Such player stations may, optionally, be integrated in a table format, may be distributed throughout a casino or other gaming site, or may include both grouped and distributed player stations.

[0093] FIG. **9** is a top view of a suitable table **1010** configured for implementing wagering games according to this disclosure. The table **1010** may include a playing surface **1404**. The table **1010** may include electronic player stations **1412**. Each player station **1412** may include a player interface **1416**, which may be used for displaying game information (e.g., graphics illustrating a

player layout, game instructions, input options, wager information, game outcomes, etc.) and accepting player elections. The player interface **1416** may be a display screen in the form of a touch screen, which may be at least substantially flush with the playing surface **1404** in some embodiments. Each player interface **1416** may be operated by its own local game processor **1414** (shown in dashed lines), although, in some embodiments, a central game processor **1428** (shown in dashed lines) may be employed and may communicate directly with player interfaces **1416**. In some embodiments, a combination of individual local game processors **1414** and the central game processor **1428** may be employed. Each of the processors **1414** and **1428** may be operably coupled to memory including one or more programs related to the rules of game play at the table **1010**.

[0094] A communication device **1460** may be included and may be operably coupled to one or more of the local game processors **1414**, the central game processor **1428**, or combinations thereof, such that information related to operation of the table **1010**, information related to the game play, or combinations thereof may be communicated between the table **1010** and other devices through a suitable communication medium, such as, for example, wired networks, Wi-Fi networks, and cellular communication networks.

[0095] The table **1010** may further include additional features, such as a dealer chip tray **1420**, which may be used by the dealer to cash players in and out of the wagering game, whereas wagers and balance adjustments during game play may be performed using, for example, virtual chips (e.g., images or text representing wagers). For embodiments using physical cards **1406a** and **1406b**, the table **1010** may further include a card-handling device **1422** such as a card shoe configured to read and deliver cards that have already been randomized. For embodiments using virtual cards, the virtual cards may be displayed at the individual player interfaces **1416**. Physical playing cards designated as “common cards” may be displayed in a common card area.

[0096] The table **1010** may further include a dealer interface **1418**, which, like the player interfaces **1416**, may include touch screen controls for receiving dealer inputs and assisting the dealer in administering the wagering game. The table **1010** may further include an upright display **1430** configured to display images that depict game information, pay tables, hand counts, historical win/loss information by player, and a wide variety of other information considered useful to the players. The upright display **1430** may be double sided to provide such information to players as well as to casino personnel.

[0097] Although an embodiment is described showing individual discrete player stations, in some embodiments, the entire playing surface **1404** may be an electronic display that is logically partitioned to permit game play from a plurality of players for receiving inputs from, and displaying game information to, the players, the dealer, or both.

[0098] FIG. **10** is a perspective view of another embodiment of a suitable electronic multi-player table **1500** configured for implementing wagering games according to the present disclosure utilizing a virtual dealer. The table **1500** may include player positions **1514** arranged in a bank about an arcuate edge **1520** of a video device **1558** that may comprise a card screen **1564** and a virtual dealer screen **1560**. The dealer screen **1560** may display a video simulation of the dealer (i.e., a virtual dealer) for interacting with the video device **1558**, such as through processing one or more stored programs stored in memory **1595** to implement the rules of game play at the video device **1558**. The dealer screen **1560** may be carried by a generally vertically extending cabinet **1562** of the video device **1558**. The substantially horizontal card screen **1564** may be configured to display at least one or more of the dealer's cards, any community cards, and each player's cards dealt by the virtual dealer on the dealer screen **1560**.

[0099] Each of the player positions **1514** may include a player interface area **1532** configured for wagering and game play interactions with the video device **1558** and virtual dealer. Accordingly, game play may be accommodated without involving physical playing cards, poker chips, and live personnel. The action may instead be simulated by a control processor **1597** interacting with and controlling the video device **1558**. The control processor **1597** may be programmed, by known

techniques, to implement the rules of game play at the video device **1558**. As such, the control processor **1597** may interact and communicate with display/input interfaces and data entry inputs for each player interface area **1532** of the video device **1558**. Other embodiments of tables and gaming devices may include a control processor that may be similarly adapted to the specific configuration of its associated device.

[0100] A communication device **1599** may be included and operably coupled to the control processor **1597** such that information related to operation of the table **1500**, information related to the game play, or combinations thereof may be communicated between the table **1500** and other devices, such as a central server, through a suitable communication medium, such as, for example, wired networks, Wi-Fi networks, and cellular communication networks.

[0101] The video device **1558** may further include banners communicating rules of play and the like, which may be located along one or more walls **1570** of the cabinet **1562**. The video device **1558** may further include additional decorative lights and speakers, which may be located on an underside surface **1566**, for example, of a generally horizontally extending top **1568** of the cabinet **1562** of the video device **1558** generally extending toward the player positions **1514**.

[0102] Although an embodiment is described showing individual discrete player stations, in some embodiments, the entire playing surface (e.g., player interface areas **1532**, card screen **1564**, etc.) may be a unitary electronic display that is logically partitioned to permit game play from a plurality of players for receiving inputs from, and displaying game information to, the players, the dealer, or both.

[0103] In some embodiments, wagering games in accordance with this disclosure may be administered using a gaming system employing a client-server architecture (e.g., over the Internet, a local area network, etc.). FIG. **11** is a schematic block diagram of an illustrative gaming system **1600** for implementing wagering games according to this disclosure. The gaming system **1600** may enable end users to remotely access game content. Such game content may include, without limitation, various types of wagering games such as card games, dice games, big wheel games, roulette, scratch off games (“scratchers”), and any other wagering game where the game outcome is determined, in whole or in part, by one or more random events. This includes, but is not limited to, Class II and Class III games as defined under 25 U.S.C. § 2701 et seq. (“Indian Gaming Regulatory Act”). Such games may include banked and/or non-banked games.

[0104] The wagering games supported by the gaming system **1600** may be operated with real currency or with virtual credits or other virtual (e.g., electronic) value indicia. For example, the real currency option may be used with traditional casino and lottery-type wagering games in which money or other items of value are wagered and may be cashed out at the end of a game session. The virtual credits option may be used with wagering games in which credits (or other symbols) may be issued to a player to be used for the wagers. A player may be credited with credits in any way allowed, including, but not limited to, a player purchasing credits; being awarded credits as part of a contest or a win event in this or another game (including non-wagering games); being awarded credits as a reward for use of a product, casino, or other enterprise, time played in one session, or games played; or may be as simple as being awarded virtual credits upon logging in at a particular time or with a particular frequency, etc. Although credits may be won or lost, the ability of the player to cash out credits may be controlled or prevented. In one example, credits acquired (e.g., purchased or awarded) for use in a play-for-fun game may be limited to non-monetary redemption items, awards, or credits usable in the future or for another game or gaming session. The same credit redemption restrictions may be applied to some or all of credits won in a wagering game as well.

[0105] An additional variation includes web-based sites having both play-for-fun and wagering games, including issuance of free (non-monetary) credits usable to play the play-for-fun games. This feature may attract players to the site and to the games before they engage in wagering. In some embodiments, a limited number of free or promotional credits may be issued to entice players

to play the games. Another method of issuing credits includes issuing free credits in exchange for identifying friends who may want to play. In another embodiment, additional credits may be issued after a period of time has elapsed to encourage the player to resume playing the game. The gaming system **1600** may enable players to buy additional game credits to allow the player to resume play. Objects of value may be awarded to play-for-fun players, which may or may not be in a direct exchange for credits. For example, a prize may be awarded or won for a highest scoring play-for-fun player during a defined time interval. All variations of credit redemption are contemplated, as desired by game designers and game hosts (the person or entity controlling the hosting systems). [0106] The gaming system **1600** may include a gaming platform to establish a portal for an end user to access a wagering game hosted by one or more gaming servers **1610** over a network **1630**. In some embodiments, games are accessed through a user interaction service **1612**. The gaming system **1600** enables players to interact with a user device **1620** through a user input device **1624** and a display **1622** and to communicate with one or more gaming servers **1610** using a network **1630** (e.g., the Internet). Typically, the user device is remote from the gaming server **1610** and the network is the word-wide web (i.e., the Internet).

[0107] In some embodiments, the gaming servers **1610** may be configured as a single server to administer wagering games in combination with the user device **1620**. In other embodiments, the gaming servers **1610** may be configured as separate servers for performing separate, dedicated functions associated with administering wagering games. Accordingly, the following description also discusses “services” with the understanding that the various services may be performed by different servers or combinations of servers in different embodiments. As shown in FIG. **11**, the gaming servers **1610** may include a user interaction service **1612**, a game service **1616**, and an asset service **1614**. In some embodiments, one or more of the gaming servers **1610** may communicate with an account server **1632** performing an account service **1632**. As explained more fully below, for some wagering type games, the account service **1632** may be separate and operated by a different entity than the gaming servers **1610**; however, in some embodiments the account service **1632** may also be operated by one or more of the gaming servers **1610**.

[0108] The user device **1620** may communicate with the user interaction service **1612** through the network **1630**. The user interaction service **1612** may communicate with the game service **1616** and provide game information to the user device **1620**. In some embodiments, the game service **1616** may also include a game engine. The game engine may, for example, access, interpret, and apply game rules. In some embodiments, a single user device **1620** communicates with a game provided by the game service **1616**, while other embodiments may include a plurality of user devices **1620** configured to communicate and provide end users with access to the same game provided by the game service **1616**. In addition, a plurality of end users may be permitted to access a single user interaction service **1612**, or a plurality of user interaction services **1612**, to access the game service **1616**. The user interaction service **1612** may enable a user to create and access a user account and interact with game service **1616**. The user interaction service **1612** may enable users to initiate new games, join existing games, and interface with games being played by the user.

[0109] The user interaction service **1612** may also provide a client for execution on the user device **1620** for accessing the gaming servers **1610**. The client provided by the gaming servers **1610** for execution on the user device **1620** may be any of a variety of implementations depending on the user device **1620** and method of communication with the gaming servers **1610**. In one embodiment, the user device **1620** may connect to the gaming servers **1610** using a web browser, and the client may execute within a browser window or frame of the web browser. In another embodiment, the client may be a stand-alone executable on the user device **1620**.

[0110] For example, the client may comprise a relatively small amount of script (e.g., JAVASCRIPT®), also referred to as a “script driver,” including scripting language that controls an interface of the client. The script driver may include simple function calls requesting information from the gaming servers **1610**. In other words, the script driver stored in the client may merely

include calls to functions that are externally defined by, and executed by, the gaming servers **1610**. As a result, the client may be characterized as a “thin client.” The client may simply send requests to the gaming servers **1610** rather than performing logic itself. The client may receive player inputs, and the player inputs may be passed to the gaming servers **1610** for processing and executing the wagering game. In some embodiments, this may involve providing specific graphical display information for the display **1622** as well as game outcomes.

[0111] As another example, the client may comprise an executable file rather than a script. The client may do more local processing than does a script driver, such as calculating where to show what game symbols upon receiving a game outcome from the game service **1616** through user interaction service **1612**. In some embodiments, portions of an asset service **1614** may be loaded onto the client and may be used by the client in processing and updating graphical displays. Some form of data protection, such as end-to-end encryption, may be used when data is transported over the network **1630**. The network **1630** may be any network, such as, for example, the Internet or a local area network.

[0112] The gaming servers **1610** may include an asset service **1614**, which may host various media assets (e.g., text, audio, video, and image files) to send to the user device **1620** for presenting the various wagering games to the end user. In other words, the assets presented to the end user may be stored separately from the user device **1620**. For example, the user device **1620** requests the assets appropriate for the game played by the user; as another example, especially relating to thin clients, just those assets that are needed for a particular display event will be sent by the gaming servers **1610**, including as few as one asset. The user device **1620** may call a function defined at the user interaction service **1612** or asset service **1614**, which may determine which assets are to be delivered to the user device **1620** as well as how the assets are to be presented by the user device **1620** to the end user. Different assets may correspond to the various user devices **1620** and their clients that may have access to the game service **1616** and to different variations of wagering games.

[0113] The gaming servers **1610** may include the game service **1616**, which may be programmed to administer wagering games and determine game play outcomes to provide to the user interaction service **1612** for transmission to the user device **1620**. For example, the game service **1616** may include game rules for one or more wagering games, such that the game service **1616** controls some or all of the game flow for a selected wagering game as well as the determined game outcomes. The game service **1616** may include pay tables and other game logic. The game service **1616** may perform random number generation for determining random game elements of the wagering game. In one embodiment, the game service **1616** may be separated from the user interaction service **1612** by a firewall or other method of preventing unauthorized access to the game service **1616** by the general members of the network **1630**.

[0114] The user device **1620** may present a gaming interface to the player and communicate the user interaction from the user input device **1624** to the gaming servers **1610**. The user device **1620** may be any electronic system capable of displaying gaming information, receiving user input, and communicating the user input to the gaming servers **1610**. For example, the user device **1620** may be a desktop computer, a laptop, a tablet computer, a set-top box, a mobile device (e.g., a smartphone), a kiosk, a terminal, or another computing device. As a specific, nonlimiting example, the user device **1620** operating the client may be an interactive electronic gaming system **1300**. The client may be a specialized application or may be executed within a generalized application capable of interpreting instructions from an interactive gaming system, such as a web browser.

[0115] The client may interface with an end user through a web page or an application that runs on a device including, but not limited to, a smartphone, a tablet, or a general computer, or the client may be any other computer program configurable to access the gaming servers **1610**. The client may be illustrated within a casino webpage (or other interface) indicating that the client is embedded into a webpage, which is supported by a web browser executing on the user device **1620**.

[0116] In some embodiments, components of the gaming system **1600** may be operated by different entities. For example, the user device **1620** may be operated by a third party, such as a casino or an individual, that links to the gaming servers **1610**, which may be operated, for example, by a wagering game service provider. Therefore, in some embodiments, the user device **1620** and client may be operated by a different administrator than the operator of the game service **1616**. In other words, the user device **1620** may be part of a third-party system that does not administer or otherwise control the gaming servers **1610** or game service **1616**. In other embodiments, the user interaction service **1612** and asset service **1614** may be operated by a third-party system. For example, a gaming entity (e.g., a casino) may operate the user interaction service **1612**, user device **1620**, or combination thereof to provide its customers access to game content managed by a different entity that may control the game service **1616**, amongst other functionality. In still other embodiments, all functions may be operated by the same administrator. For example, a gaming entity (e.g., a casino) may elect to perform each of these functions in-house, such as providing access to the user device **1620**, delivering the actual game content, and administering the gaming system **1600**.

[0117] The gaming servers **1610** may communicate with one or more external account servers **1632** (also referred to herein as an account service **1632**), optionally through another firewall. For example, the gaming servers **1610** may not directly accept wagers or issue payouts. That is, the gaming servers **1610** may facilitate online casino gaming but may not be part of self-contained online casino itself. Another entity (e.g., a casino or any account holder or financial system of record) may operate and maintain its external account service **1632** to accept bets and make payout distributions. The gaming servers **1610** may communicate with the account service **1632** to verify the existence of funds for wagering and to instruct the account service **1632** to execute debits and credits. As another example, the gaming servers **1610** may directly accept bets and make payout distributions, such as in the case where an administrator of the gaming servers **1610** operates as a casino.

[0118] Additional features may be supported by the gaming servers **1610**, such as hacking and cheating detection, data storage and archival, metrics generation, messages generation, output formatting for different end user devices, as well as other features and operations.

[0119] FIG. **12** is a schematic block diagram of a table **1682** for implementing wagering games including a live dealer video feed. Features of the gaming system **1600** (see FIG. **11**) described above in connection with FIG. **11** may be utilized in connection with this embodiment, except as further described. Rather than cards being determined by computerized random processes, physical cards (e.g., from a standard, 52-card deck of playing cards) may be dealt by a live dealer **1680** at a table **1682** from a card-handling system **1684** located in a studio or on a casino floor. A table manager **1686** may assist the dealer **1680** in facilitating play of the game by transmitting a live video feed of the dealer's actions to the user device **1620** and transmitting remote player elections to the dealer **1680**. As described above, the table manager **1686** may act as or communicate with a gaming system **1600** (see FIG. **11**) (e.g., acting as the gaming system **1600** (see FIG. **11**) itself or as an intermediate client interposed between and operationally connected to the user device **1620** and the gaming system **1600** (see FIG. **11**)) to provide gaming at the table **1682** to users of the gaming system **1600** (see FIG. **11**). Thus, the table manager **1686** may communicate with the user device **1620** through a network **1630** (see FIG. **11**), and may be a part of a larger online casino, or may be operated as a separate system facilitating game play. In various embodiments, each table **1682** may be managed by an individual table manager **1686** constituting a gaming device, which may receive and process information relating to that table. For simplicity of description, these functions are described as being performed by the table manager **1686**, though certain functions may be performed by an intermediary gaming system **1600** (see FIG. **11**), such as the one shown and described in connection with FIG. **11**. In some embodiments, the gaming system **1600** (see FIG. **11**) may match remotely located players to tables **1682** and facilitate transfer of information between

user devices **1620** and tables **1682**, such as wagering amounts and player option elections, without managing gameplay at individual tables. In other embodiments, functions of the table manager **1686** may be incorporated into a gaming system **1600** (sec FIG. **11**).

[0120] The table **1682** includes a camera **1670** and optionally a microphone **1672** to capture video and audio feeds relating to the table **1682**. The camera **1670** may be trained on the live dealer **1680**, play area **1687**, and card-handling system **1684**. As the game is administered by the live dealer **1680**, the video feed captured by the camera **1670** may be shown to the player remotely using the user device **1620**, and any audio captured by the microphone **1672** may be played to the player remotely using the user device **1620**. In some embodiments, the user device **1620** may also include a camera, microphone, or both, which may also capture feeds to be shared with the dealer **1680** and other players. In some embodiments, the camera **1670** may be trained to capture images of the card faces, chips, and chip stacks on the surface of the gaming table. Known image extraction techniques may be used to obtain card count and card rank and suit information from the card images.

[0121] Card and wager data in some embodiments may be used by the table manager **1686** to determine game outcome. The data extracted from the camera **1670** may be used to confirm the card data obtained from the card-handling system **1684**, to determine a player position that received a card, and for general security monitoring purposes, such as detecting player or dealer card switching, for example. Examples of card data include, for example, suit and rank information of a card, suit and rank information of each card in a hand, rank information of a hand, and rank information of every hand in a round of play.

[0122] The live video feed permits the dealer to show cards dealt by the card-handling system **1684** and play the game as though the player were at a gaming table, playing with other players in a live casino. In addition, the dealer can prompt a user by announcing a player's election is to be performed. In embodiments where a microphone **1672** is included, the dealer **1680** can verbally announce action or request an election by a player. In some embodiments, the user device **1620** also includes a camera or microphone, which also captures feeds to be shared with the dealer **1680** and other players.

[0123] The card-handling system **1684** may be as shown and was described previously. The play area **1686** depicts player layouts for playing the game. As determined by the rules of the game, the player at the user device **1620** may be presented options for responding to an event in the game using a client as described with reference to FIG. **11**.

[0124] Player elections may be transmitted to the table manager **1686**, which may display player elections to the dealer **1680** using a dealer display **1688** and player action indicator **1690** on the table **1682**. For example, the dealer display **1688** may display information regarding where to deal the next card or which player position is responsible for the next action.

[0125] In some embodiments, the table manager **1686** may receive card information from the card-handling system **1684** to identify cards dealt by the card-handling system **1684**. For example, the card-handling system **1684** may include a card reader to determine card information from the cards. The card information may include the rank and suit of each dealt card and hand information.

[0126] The table manager **1686** may apply game rules to the card information, along with the accepted player decisions, to determine gameplay events and wager results. Alternatively, the wager results may be determined by the dealer **1680** and input to the table manager **1686**, which may be used to confirm automatically determined results by the gaming system.

[0127] Card and wager data in some embodiments may be used by the table manager **1686** to determine game outcome. The data extracted from the camera **1670** may be used to confirm the card data obtained from the card-handling system **1684**, to determine a player position that received a card, and for general security monitoring purposes, such as detecting player or dealer card switching, for example.

[0128] The live video feed permits the dealer to show cards dealt by the card-handling system **1684**

and play the game as though the player were at a live casino. In addition, the dealer can prompt a user by announcing a player's election is to be performed. In embodiments where a microphone **1672** is included, the dealer **1680** can verbally announce action or request an election by a player. In some embodiments, the user device **1620** also includes a camera or microphone, which also captures feeds to be shared with the dealer **1680** and other players.

[0129] FIG. **13** is a simplified block diagram showing elements of computing devices that may be used in systems and apparatuses of this disclosure. A computing system **1640** may be a user-type computer, a file server, a computer server, a notebook computer, a tablet, a handheld device, a mobile device, or other similar computer system for executing software. The computing system **1640** may be configured to execute software programs containing computing instructions and may include one or more processors **1642**, memory **1646**, one or more displays **1658**, one or more user interface elements **1644**, one or more communication elements **1656**, and one or more storage devices **1648** (also referred to herein simply as storage **1648**).

[0130] The processors **1642** may be configured to execute a wide variety of operating systems and applications including the computing instructions for administering wagering games of the present disclosure.

[0131] The processors **1642** may be configured as a general-purpose processor such as a microprocessor, but in the alternative, the general-purpose processor may be any processor, controller, microcontroller, or state machine suitable for carrying out processes of the present disclosure. The processor **1642** may also be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0132] A general-purpose processor may be part of a general-purpose computer. However, when configured to execute instructions (e.g., software code) for carrying out embodiments of the present disclosure the general-purpose computer should be considered a special-purpose computer. Moreover, when configured according to embodiments of the present disclosure, such a special-purpose computer improves the function of a general-purpose computer because, absent the present disclosure, the general-purpose computer would not be able to carry out the processes of the present disclosure. The processes of the present disclosure, when carried out by the special-purpose computer, are processes that a human would not be able to perform in a reasonable amount of time due to the complexities of the data processing, decision making, communication, interactive nature, or combinations thereof for the present disclosure. The present disclosure also provides meaningful limitations in one or more particular technical environments that go beyond an abstract idea. For example, embodiments of the present disclosure provide improvements in the technical field related to the present disclosure.

[0133] The memory **1646** may be used to hold computing instructions, data, and other information for performing a wide variety of tasks including administering wagering games of the present disclosure. By way of example, and not limitation, the memory **1646** may include Synchronous Random Access Memory (SRAM), Dynamic RAM (DRAM), Read-Only Memory (ROM), Flash memory, and the like.

[0134] The display **1658** may be a wide variety of displays such as, for example, light-emitting diode displays, liquid crystal displays, cathode ray tubes, and the like. In addition, the display **1658** may be configured with a touch-screen feature for accepting user input as a user interface element **1644**.

[0135] As nonlimiting examples, the user interface elements **1644** may include elements such as displays, keyboards, push-buttons, mice, joysticks, haptic devices, microphones, speakers, cameras, and touchscreens.

[0136] As nonlimiting examples, the communication elements **1656** may be configured for communicating with other devices or communication networks. As nonlimiting examples, the communication elements **1656** may include elements for communicating on wired and wireless

communication media, such as for example, serial ports, parallel ports, Ethernet connections, universal serial bus (USB) connections, IEEE 1394 (“firewire”) connections, THUNDERBOLT™ connections, BLUETOOTH® wireless networks, ZigBee wireless networks, 802.11 type wireless networks, cellular telephone/data networks, fiber optic networks and other suitable communication interfaces and protocols.

[0137] The storage **1648** may be used for storing relatively large amounts of nonvolatile information for use in the computing system **1640** and may be configured as one or more storage devices. By way of example and not limitation, these storage devices may include computer-readable media (CRM). This CRM may include, but is not limited to, magnetic and optical storage devices such as disk drives, magnetic tape, CDs (compact discs), DVDs (digital versatile discs or digital video discs), and semiconductor devices such as RAM, DRAM, ROM, EPROM, Flash memory, and other equivalent storage devices.

[0138] A person of ordinary skill in the art will recognize that the computing system **1640** may be configured in many different ways with different types of interconnecting buses between the various elements. Moreover, the various elements may be subdivided physically, functionally, or a combination thereof. As one nonlimiting example, the memory **1646** may be divided into cache memory, graphics memory, and main memory. Each of these memories may communicate directly or indirectly with the one or more processors **1642** on separate buses, partially combined buses, or a common bus.

[0139] As a specific, nonlimiting example, various methods and features of the present disclosure may be implemented in a mobile, remote, or mobile and remote environment over one or more of Internet, cellular communication (e.g., Broadband), near field communication networks and other communication networks referred to collectively herein as an iGaming environment. The iGaming environment may be accessed through social media environments such as FACEBOOK® and the like. DragonPlay Ltd, acquired by Bally Technologies Inc., provides an example of a platform to provide games to user devices, such as cellular telephones and other devices utilizing ANDROID®, IPHONE® and FACEBOOK® platforms. Where permitted by jurisdiction, the iGaming environment can include pay-to-play (P2P) gaming where a player, from their device, can make value based wagers and receive value based awards. Where P2P is not permitted the features can be expressed as entertainment only gaming where players wager virtual credits having no value or risk no wager whatsoever such as playing a promotion game or feature.

[0140] FIG. **14** illustrates an illustrative embodiment of information flows in an iGaming environment. At a player level, the player or user accesses a site hosting the activity such as a website **1700**. The website **1700** may functionally provide a web game client **1702**. The web game client **1702** may be, for example, represented by a game client **1708** downloadable at information flow **1710**, which may process applets transmitted from a gaming server **1714** at information flow **1711** for rendering and processing game play at a player's remote device. Where the game is a P2P game, the gaming server **1714** may process value-based wagers (e.g., money wagers) and randomly generate an outcome for rendition at the player's device. In some embodiments, the web game client **1702** may access a local memory store to drive the graphic display at the player's device. In other embodiments, all or a portion of the game graphics may be streamed to the player's device with the web game client **1702** enabling player interaction and display of game features and outcomes at the player's device.

[0141] The website **1700** may access a player-centric, iGaming-platform-level account module **1704** at information flow **1706** for the player to establish and confirm credentials for play and, where permitted, access an account (e.g., an eWallet) for wagering. The account module **1704** may include or access data related to the player's profile (e.g., player-centric information desired to be retained and tracked by the host), the player's electronic account, deposit, and withdrawal records, registration and authentication information, such as username and password, name and address information, date of birth, a copy of a government issued identification document, such as a driver's

license or passport, and biometric identification criteria, such as fingerprint or facial recognition data, and a responsible gaming module containing information, such as self-imposed or jurisdictionally imposed gaming restraints, such as loss limits, daily limits and duration limits. The account module **1704** may also contain and enforce geo-location limits, such as geographic areas where the player may play P2P games, user device IP address confirmation, and the like.

[0142] The account module **1704** communicates at information flow **1705** with a game module **1716** to complete log-ins, registrations, and other activities. The game module **1716** may also store or access a player's gaming history, such as player tracking and loyalty club account information. The game module **1716** may provide static web pages to the player's device from the game module **1716** through information flow **1718**, whereas, as stated above, the live game content may be provided from the gaming server **1714** to the web game client through information flow **1711**.

[0143] The gaming server **1714** may be configured to provide interaction between the game and the player, such as receiving wager information, game selection, inter-game player selections or choices to play a game to its conclusion, and the random selection of game outcomes and graphics packages, which, alone or in conjunction with the downloadable game client **1708**/web game client **1702** and game module **1716**, provide for the display of game graphics and player interactive interfaces. At information flow **1718**, player account and log-in information may be provided to the gaming server **1714** from the account module **1704** to enable gaming. Information flow **1720** provides wager/credit information between the account module **1704** and gaming server **1714** for the play of the game and may display credits and eWallet availability. Information flow **1722** may provide player tracking information for the gaming server **1714** for tracking the player's play. The tracking of play may be used for purposes of providing loyalty rewards to a player, determining preferences, and the like.

[0144] All or portions of the features of FIG. **14** may be supported by servers and databases located remotely from a player's mobile device and may be hosted or sponsored by regulated gaming entity for P2P gaming or, where P2P is not permitted, for entertainment only play.

[0145] In some embodiments, wagering games may be administered in an at least partially player-pooled format, with payouts on pooled wagers being paid from a pot to players and losses on wagers being collected into the pot and eventually distributed to one or more players. Such player-pooled embodiments may include a player-pooled progressive embodiment, in which a pot is eventually distributed when a predetermined progressive-winning hand combination or composition is dealt. Player-pooled embodiments may also include a dividend refund embodiment, in which at least a portion of the pot is eventually distributed in the form of a refund distributed, e.g., pro-rata, to the players who contributed to the pot.

[0146] In some player-pooled embodiments, the game administrator may not obtain profits from chance-based events occurring in the wagering games that result in lost wagers. Instead, lost wagers may be redistributed back to the players. To profit from the wagering game, the game administrator may retain a commission, such as, for example, a player entrance fee or a rake taken on wagers, such that the amount obtained by the game administrator in exchange for hosting the wagering game is limited to the commission and is not based on the chance events occurring in the wagering game itself. The game administrator may also charge a rent of flat fee to participate.

[0147] It is noted that the methods described herein can be played with any number of standard decks of 52 cards (e.g., 1 deck to 10 decks). A standard deck is a collection of cards comprising an Ace, two, three, four, five, six, seven, eight, nine, ten, jack, queen, king, for each of four suits (comprising spades, diamonds, clubs, hearts) totaling 52 cards. Cards can be shuffled or a continuous shuffling machine (CSM) can be used. A standard deck of 52 cards can be used, as well as other kinds of decks, such as Spanish decks, decks with wild cards, etc. The operations described herein can be performed in any sensible order. Furthermore, numerous different variants of house rules can be applied.

[0148] Note that in the embodiments played using computers (a processor/processing unit), “virtual

deck(s)” of cards are used instead of physical decks. A virtual deck is an electronic data structure used to represent a physical deck of cards which uses electronic representations for each respective card in the deck. In some embodiments, a virtual card is presented (e.g., displayed on an electronic output device using computer graphics, projected onto a surface of a physical table using a video projector, etc.) and is presented to mimic a real life image of that card.

[0149] Methods described herein can also be played on a physical table using physical cards and physical chips used to place wagers. Such physical chips can be directly redeemable for cash. When a player wins (dealer loses) the player's wager, the dealer will pay that player a respective payout amount. When a player loses (dealer wins) the player's wager, the dealer will take (collect) that wager from the player and typically place those chips in the dealer's chip rack. All rules, embodiments, features, etc. of a game being played can be communicated to the player (e.g., verbally or on a written rule card) before the game begins.

[0150] Initial cash deposits can be made into the electronic gaming machine which converts cash into electronic credits. Wagers can be placed in the form of electronic credits, which can be cashed out for real coins or a ticket (e.g., ticket-in-ticket-out) which can be redeemed at a casino cashier or kiosk for real cash and/or coins.

[0151] Any component of any embodiment described herein may include hardware, software, or any combination thereof.

[0152] Further, the operations described herein can be performed in any sensible order. Any operations not required for proper operation can be optional. Further, all methods described herein can also be stored as instructions on a computer readable storage medium, which instructions are operable by a computer processor. All variations and features described herein can be combined with any other features described herein without limitation. All features in all documents incorporated by reference herein can be combined with any feature(s) described herein, and also with all other features in all other documents incorporated by reference, without limitation.

[0153] Features of various embodiments of the inventive subject matter described herein, however essential to the example embodiments in which they are incorporated, do not limit the inventive subject matter as a whole, and any reference to the invention, its elements, operation, and application are not limiting as a whole, but serve only to define these example embodiments. This detailed description does not, therefore, limit embodiments which are defined only by the appended claims. Further, since numerous modifications and changes may readily occur to those skilled in the art, it is not desired to limit the inventive subject matter to the exact construction and operation illustrated and described, and accordingly all suitable modifications and equivalents may be resorted to, falling within the scope of the inventive subject matter.

Claims

1. An apparatus comprising: a token tray having a plurality of light sensors, wherein the plurality of light sensors are positioned inside a token support structure of the token tray, the token support structure having a structure that positions a group of gaming tokens into an orientation from which a respective edge of each gaming token can be viewed by an image sensor; and a tracking controller configured to perform operations that cause the apparatus to: detect a level of ambient light at each of the plurality of light sensors; determine, in response to detection of the level of ambient light at each of the plurality of light sensors, a number of gaming tokens placed inside the token support structure; identify, by a machine learning model via electronic analysis of image data of the gaming tokens inside the token support structure, a first pattern on the edge of at least a relative majority of the gaming tokens, and a second, different pattern on at least one minority gaming token; determine, based on the first pattern, a primary token denomination value associated with the token support structure; determine, by the machine learning model based on the second pattern, a specific denomination value for the at least one minority gaming token; and indicate, via

an electronic communication to one or more electronic devices, the presence of the at least one minority gaming token along with its specific denomination value being different from the primary token denomination.

2. The apparatus of claim 1, wherein the token support structure comprises a sloped column or a semi-cylindrical slot configured to group circular-shaped gaming tokens into a stack.

3. The apparatus of claim 1, wherein the plurality of light sensors comprise at least one of light dependent resistors (LDRs) or photodiode-based proximity sensors.

4. The apparatus of claim 1, further comprising a calibration module configured to dynamically adjust an operational parameter of at least one of the light sensors or the machine learning model based on at least one of a monitored environmental parameter or a determined level of pattern mismatch in gaming token.

5. The apparatus of claim 1, further comprising a diagnostic module configured to assess an operational status of the plurality of light sensors and report anomalies.

6. The apparatus of claim 1, wherein the machine learning model comprises analysis logic configured to improve identification accuracy for varied gaming token edge conditions, including worn or soiled gaming token edges.

7. The apparatus of claim 1, wherein the token support structure is associated with a local microcontroller for performing at least the detecting and the determining steps for its respective token support structure, the local microcontroller communicatively coupled to the tracking controller which acts as a central controller for aggregating data and performing the identifying, determining denomination values, and indicating steps.

8. The apparatus of claim 1, wherein the tracking controller is further configured to: compute a token-grouping value for the token support structure, wherein the token-grouping value is calculated by summing: (i) a value derived from the number of gaming tokens exhibiting the first pattern multiplied by the primary token denomination value, and (ii) a value derived from the at least one minority gaming token multiplied by its specific denomination value; and indicate, in response to communication with the one or more electronic devices, the token-grouping value.

9. A method of tracking gaming tokens using an apparatus that includes a token tray having a plurality of light sensors positioned inside a token support structure of the token tray, the token support structure having a structure that positions a group of gaming tokens into an orientation from which a respective edge of each gaming token can be viewed by an image sensor, and a tracking controller, the method comprising: detecting, by the tracking controller using the plurality of light sensors, a level of ambient light at each of the plurality of light sensors; determining, by the tracking controller in response to detection of the level of ambient light at each of the plurality of light sensors, a number of gaming tokens placed inside the token support structure; identifying, by the tracking controller using a machine learning model via electronic analysis of image data of the gaming tokens inside the token support structure, a first pattern on the edge of at least a relative majority of the gaming tokens, and a second, different pattern on at least one minority gaming token; determining, by the tracking controller based on the first pattern, a primary token denomination value associated with the token support structure; determining, by the tracking controller using the machine learning model and based on the second pattern, a specific denomination value for the at least one minority gaming token; and indicating, by the tracking controller via an electronic communication to one or more electronic devices, the presence of the at least one minority gaming token along with its specific denomination value being different from the primary token denomination.

10. The method of claim 9, wherein the token support structure of the apparatus comprises a sloped column or a semi-cylindrical slot configured to group circular-shaped gaming tokens into a stack.

11. The method of claim 9, wherein the plurality of light sensors comprise at least one of light dependent resistors (LDRs) or photodiode-based proximity sensors.

12. The method of claim 9, wherein the apparatus further comprises a calibration module, and the

method further comprises dynamically adjusting, by the tracking controller using the calibration module, an operational parameter of at least one of the light sensors or the machine learning model based on at least one of a monitored environmental parameter or a determined level of pattern mismatch in gaming token identification.

13. The method of claim 9, wherein the apparatus further comprises a diagnostic module, and the method further comprises assessing, by the tracking controller using the diagnostic module, an operational status of the plurality of light sensors and reporting anomalies.

14. The method of claim 9, wherein the machine learning model used in the identifying step comprises advanced analysis logic configured to improve identification accuracy for varied gaming token edge conditions, including worn or soiled gaming token edges.

15. The method of claim 9, wherein the token support structure of the apparatus is associated with a local microcontroller, and wherein at least the detecting and the determining steps for the respective token support structure are performed by its associated local microcontroller, and further comprising communicating data from the local microcontroller to the tracking controller for performing the identifying, determining denomination values, and indicating steps.

16. The method of claim 9, further comprising: computing, by the tracking controller, a token-grouping value for the token support structure by summing: (i) a value derived from the number of gaming tokens exhibiting the first pattern multiplied by the primary token denomination value, and (ii) a value derived from the at least one minority gaming token multiplied by its specific denomination value; and indicating, by the tracking controller in response to communication with the one or more electronic devices, the token-grouping value.

17. One or more non-transitory machine-readable mediums having instructions stored thereon, which when executed by a set of one or more processors of an apparatus cause the apparatus to perform operations comprising: detecting a level of ambient light at each of a plurality of light sensors of a chip tray, wherein the plurality of light sensors are positioned inside a column of the chip tray; determining, in response to detection of the level of ambient light at each of the plurality of light sensors, a number of gaming chips placed inside the column; identifying, by a machine learning model via electronic analysis of image data of the gaming chips inside the column, a first pattern on the edge of at least a relative majority of the gaming chips, and a second, different pattern on at least one minority gaming chip; determining, based on the first pattern, a primary chip denomination value associated with the column; determining, by the machine learning model based on the second pattern, a specific denomination value for the at least one minority gaming chip; and indicating, via an electronic communication to one or more electronic devices, the presence of the at least one minority gaming chip along with its specific denomination value being different from the primary chip denomination.
